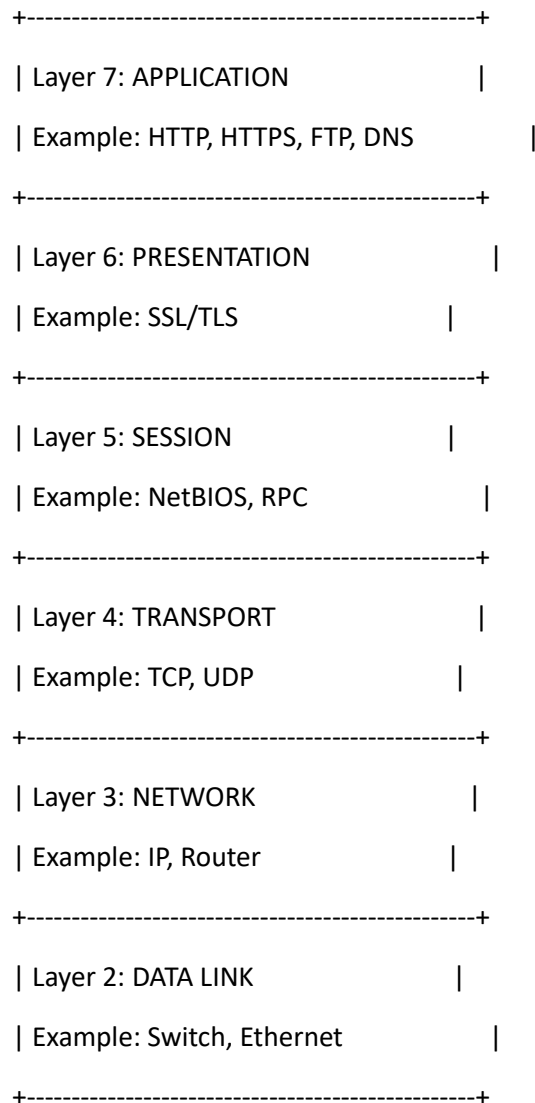


N+ - Network Fundamentals and building network

OSI Layer

[Q-1] Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer (e.g., Switch at Data link, HTTP at Application)

OSI 7-LAYER MODEL



| Layer 1: PHYSICAL |

| Example: Hub, Repeater, UTP/Fiber Cable |

+-----+

Data Flow

Sender

|

v

+-----+

| Application |

+-----+

| Presentation |

+-----+

| Session |

+-----+

| Transport |

+-----+

| Network |

+-----+

| Data Link |

+-----+

| Physical |

+-----+

|

v

Transmission Medium

|

v

Receiver

|

v



[Q-2] Match the following networking scenarios to the correct OSI layer: a) Encrypting data before sending, b) Routing packets across networks, c) Converting data to electrical signals, d) Opening a web page in your browser, e) Detecting and correcting errors in frames.

Hint:
Write your answers as 'Scenario: Layer Name (Layer Number)'.

a) Encrypting data before sending: Presentation Layer (Layer 6)

b) Routing packets across networks: Network Layer (Layer 3)

c) Converting data to electrical signals: Physical Layer (Layer 1)

d) Opening a web page in your browser: Application Layer (Layer 7)

e) Detecting and correcting errors in frames: Data Link Layer (Layer 2)

[Q-3] Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.

Example: WhatsApp Message Sending

When I send a message on WhatsApp, several OSI layers work together:

1. Application Layer (Layer 7)

WhatsApp provides the user interface where I type and send a message.

The app creates the message data and sends it for transmission.

2. Presentation Layer (Layer 6)

WhatsApp encrypts the message using end-to-end encryption before sending.

This layer ensures the data is secure and readable only by the intended recipient.

3. Transport Layer (Layer 4)

Protocols such as TCP ensure the message is delivered reliably.

The data is divided into segments and checked for errors during transmission.

4. Network Layer (Layer 3)

IP addresses are used to route the message from my phone to WhatsApp servers and then to the receiver's device.

Routers help forward the packets across different networks.

5. Data Link Layer (Layer 2)

Wi-Fi or mobile network technology uses MAC addresses to transfer frames between devices on the local network.

Error detection is performed on frames.

6. Physical Layer (Layer 1)

The message is transmitted as electrical, radio, or optical signals through Wi-Fi, mobile towers, cables, or fibre networks.

[Q-4] Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning.

Network Layer (Layer 3) is working correctly because ICMP ping packets can reach the destination and return.

Data Link Layer (Layer 2) and Physical Layer (Layer 1) are also functioning, since communication is occurring.

The problem is likely at the higher OSI layers:

1. Application Layer (Layer 7)

The web service (HTTP/HTTPS) may be down.

The web server may not be running.

The website may be configured incorrectly.

DNS issues could occur if accessing by name instead of IP.

2. Transport Layer (Layer 4)

TCP ports required for web traffic may be blocked.

HTTP uses Port 80.

HTTPS uses Port 443.

A firewall may be blocking TCP connections while still allowing ICMP ping packets.

Reasoning

Ping uses ICMP, which operates at the Network Layer, whereas web browsing uses HTTP/HTTPS at the Application Layer and TCP at the Transport Layer. Since ping works but the website does not open, the lower layers are functioning correctly, and the issue is most likely at the Transport Layer (Layer 4) or Application Layer (Layer 7).

Answer: The problem is likely in the Transport Layer (Layer 4) or Application Layer (Layer 7) because network connectivity exists (ping succeeds), but the web service cannot be accessed.